

|          |                                                                                  |          |
|----------|----------------------------------------------------------------------------------|----------|
| <b>1</b> | From the given equation,<br>$\mu^2 = gL^3 \rho_l (\rho - \rho_l) / Ar$           | <b>1</b> |
|          | $[\mu^2] = (\text{m s}^{-2}) (\text{m})^3 (\text{kg m}^{-3}) (\text{kg m}^{-3})$ | <b>1</b> |
|          | Thus $[\mu] = \text{kg m}^{-1} \text{s}^{-1}$                                    | <b>1</b> |

|          |             |                                                                                 |          |
|----------|-------------|---------------------------------------------------------------------------------|----------|
| <b>2</b> | <b>(a)</b>  | The point of a body through which the entire weight of the body appears to act. | <b>1</b> |
|          | <b>(i)</b>  |                                                                                 |          |
|          | <b>(ii)</b> | The force between two surfaces that opposes relative motion between them.       | <b>1</b> |

|   |     |      |                                                                                                                                                                 |                     |
|---|-----|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 2 | (b) | (i)  | 1. $x = 78 - 42 = 36 \text{ m}$<br><br>2. $\text{EPE} = \frac{1}{2} Fx = \frac{1}{2} (3800)(36)$<br><br>$= 68400 \text{ J}$                                     | 1<br><br>1<br><br>1 |
|   | (b) | (ii) | GPE lost = EPE gained or a statement to the same effect<br><br>$Mgh = 68400$<br><br>$M = 68400 / (9.81 \times 78) = 89.4 \text{ kg}$<br>$\approx 90 \text{ kg}$ | 1<br><br>1          |

|   |     |     |                                                                        |  |
|---|-----|-----|------------------------------------------------------------------------|--|
| 2 | (a) | (i) | $nM = (1/2) N m \langle \sigma^2 \rangle$ , where $\sigma = N m / \mu$ |  |
|---|-----|-----|------------------------------------------------------------------------|--|